CO2Flux: Relative Spectral Phases of Subsurface CO_2 at Biosphere 2 LEO

A Unified Report of Documented Results and Open Validation

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19 October 2026

Abstract

The current approved task is to display and evaluate the temporal evolution of three adjacent relative Fourier phases for a selected West-slope four-channel CO_2 profile, from atmospheric air to 5, 20 and 50 cm depth. Existing preliminary generalized Lomb–Scargle amplitude spectra and a separate direct-Fourier pilot are reported without conflating their estimators. The saved direct-Fourier analysis produces three finite phase coordinates in 492 overlapping 30-day windows stepped by one day at one cycle per day. All 492 windows contain unresolved sensor-review flags; their scientific reliability is not established. An isolated 5-to-20 cm phase change from 17 to 18 December 2025 is diagnosed algebraically but not physically attributed. Earlier studies of the times of maxima of fitted daily harmonics remain historical and do not answer the new spectral-phase question. Neither the final irrigation calendar nor I. R. Gabitov's Stage 0.5 methodological approval has been recorded as complete.

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Research roadmap and source correspondence

The identifiers below follow the existing research README and the approved PLAN. They identify research stages, not the numbered sections of this report. In particular, **0.5/Q1** is a completed diagnostic of one phase change, *not* completion of Stage 0.5. Existing numbered historical notebooks concern the previous methodology, not the future spectral chapters.

Stage	Object and status	Report section	Canonical source
0.1	Irrigation calendar; incomplete	Limitations	PLAN: 0.1
0.2	Data and quality control; science-level admission unresolved	Data and QC	Data checks
0.3	Preliminary amplitude spectra; calculated, not a final frequency choice	Amplitude spectra	Spectra notebook
0.4	Relative Fourier phases; 492 computed windows, reliability unestablished	Direct Fourier; phase series	Phase notebook, sections 1–9
0.5/Q1	One phase jump analyzed; does not close Stage 0.5	Single-jump result	Phase notebook, section 10
0.5	Presentation to I.G. and explicit approval; pending	Next approved steps	PLAN: 0.5 and stopping criteria
1–3	Future spectral-phase chapters; not yet begun	Next approved steps	PLAN: later stages
4	Buchstaber method; methodological preparation only	Buchstaber	Buchstaber notebook
5–6	Spatial links and physical mechanisms; future work	Next approved steps	PLAN: later stages
Historic 1–3	Absolute-harmonic maximum times/amplitudes; distinct earlier work	Historical analyses	Notebook 01, 02, 03

1 Introduction and research status

The long-term objective is to assess whether changes in CO_2 accompanying the transformation of crushed basalt into soil allow transport, chemical interactions and biological sources or sinks to be distinguished. The immediate approved observable is instead precisely defined: three relative spectral phases between consecutive levels. Absolute times of fitted diurnal maxima are not interchangeable with these quantities. The current canonical README and PLAN place the project at preparatory Stage 0, before the new Chapter 1 [1, 2]. This report assembles existing results, not new scientific computations or an alternative research plan.

2 LEO experimental setup

The selected pilot vertical on LEO West is at $(x, y) = (1, 4)$ m. Four channels are used: air, 5, 20 and 50 cm. The air LI-COR instrument is at $(0, 4)$ m, horizontally 1 m from the basalt profile. Its nominal height +0.25 m comes from Excel metadata and is not independently established by the Oracle BOXZ field. The assumption $C_0(t) = C_{\mathrm{air}}(t)$ is adopted for the present analysis and must not be mistaken for experimental proof of an exact surface boundary condition [4, 5].

Level	Sensor	Model	Stored unit
Air	1275	LI-COR	μmol/mol
5 cm	996	GMM222	ppm
20 cm	1012	GMM222	ppm
50 cm	1035	GMM222	ppm

The historical West-slope profile W_R4_C-4 is different: sensors 994/1010/1026 at 5/20/35 cm. Its results are not pooled with this four-channel 5/20/50 cm pilot.

3 Data and quality control

3.1 Input and temporal coverage

The versioned archive is `20261002T061358Z-264980`. The pilot calendar is the half-open interval [2025-04-29, 2026-10-02), containing 521 days; 520 have some four-channel daily overlap. The raw atmospheric record has 7,589 measurements; each selected subsurface sensor has 50,009 raw measurements, with 49,784, 49,770 and 49,715 technically retained measurements, respectively. Median native intervals are about two hours for air and 15 minutes for the basalt channels. The database time base is taken as confirmed: no additional time shift is imposed [3, 5].

The air channel has 7,589 records flagged because its LI-COR artifact and range checks remain unconfirmed. The corresponding subsurface counts marked for ambiguous excursions are 188, 193 and 206. A technically retained value can still be doubtful. Previously established subsurface instrument thresholds do not automatically apply to LI-COR, and old monthly amplitude criteria are not universal spectral-admission rules [2, 8].

3.2 Common two-hour cells and missingness

Let R_{jk} be retained observations for channel j within $B_k = [s_k, s_k + 2\,\mathrm{h})$. Only when $n_{jk} = |R_{jk}| > 0$ define the bin mean

$$x_{jk} = \frac{1}{n_{jk}} \sum_{r \in R_{jk}} C_{jr}. \tag{1}$$

Each bin retains both its count and the logical union of review flags. For a window W, set

$$I_W = \{k : B_k \subset W,\ n_{jk} > 0 \text{ for every } j = 0, 1, 2, 3\}, \qquad N = |I_W|. \tag{2}$$

All four estimates use these same actually observed bin centers, without interpolating, zero filling, compressing the calendar or carrying the last air value forward. The complete pilot calendar contains 6,184 common two-hour bins out of 6,252 possible. For the saved 30-day windows there are 317–360 common cells per window and 88.06–100% coverage; the maximum common-bin gap is 42 hours. No new science-level minimum coverage or amplitude threshold has been approved [6].

4 Fourier methodology

4.1 Stage 0.3: amplitude spectra

The saved amplitude spectra use a floating-intercept, equal-weight generalized Lomb–Scargle sinusoidal least-squares model at each candidate frequency:

$$C_j(t) \simeq c_j(f) + a_j(f)\cos(2\pi f t) + b_j(f)\sin(2\pi f t) = c_j(f) + \mathrm{Re}\{Z_j(f)e^{2\pi i f t}\}, \quad Z_j = a_j - ib_j, \quad A_j(f) = |Z_j|. \tag{3}$$

A_j is a fitted sinusoidal semi-amplitude in the source unit, not spectral power. This is not the direct Fourier estimator employed in Stage 0.4. The existing method is implemented by `gls(t,y,frequency)` and `grid(days)` in `Project_description/sensorDB/preliminary_spectra.py`. The frequency step is $1/T \simeq 0.001921153$ cycles/day for an observed span $T \simeq 520.5208$ days, over $0 < f < 6$ cycles/day. The display bound is not a strict Nyquist claim for irregular sampling. Other spectral peaks are documented, but their mechanisms and an optimal physical frequency have not been established. The 24-hour harmonic is merely the approved provisional demonstration frequency [3, 8].

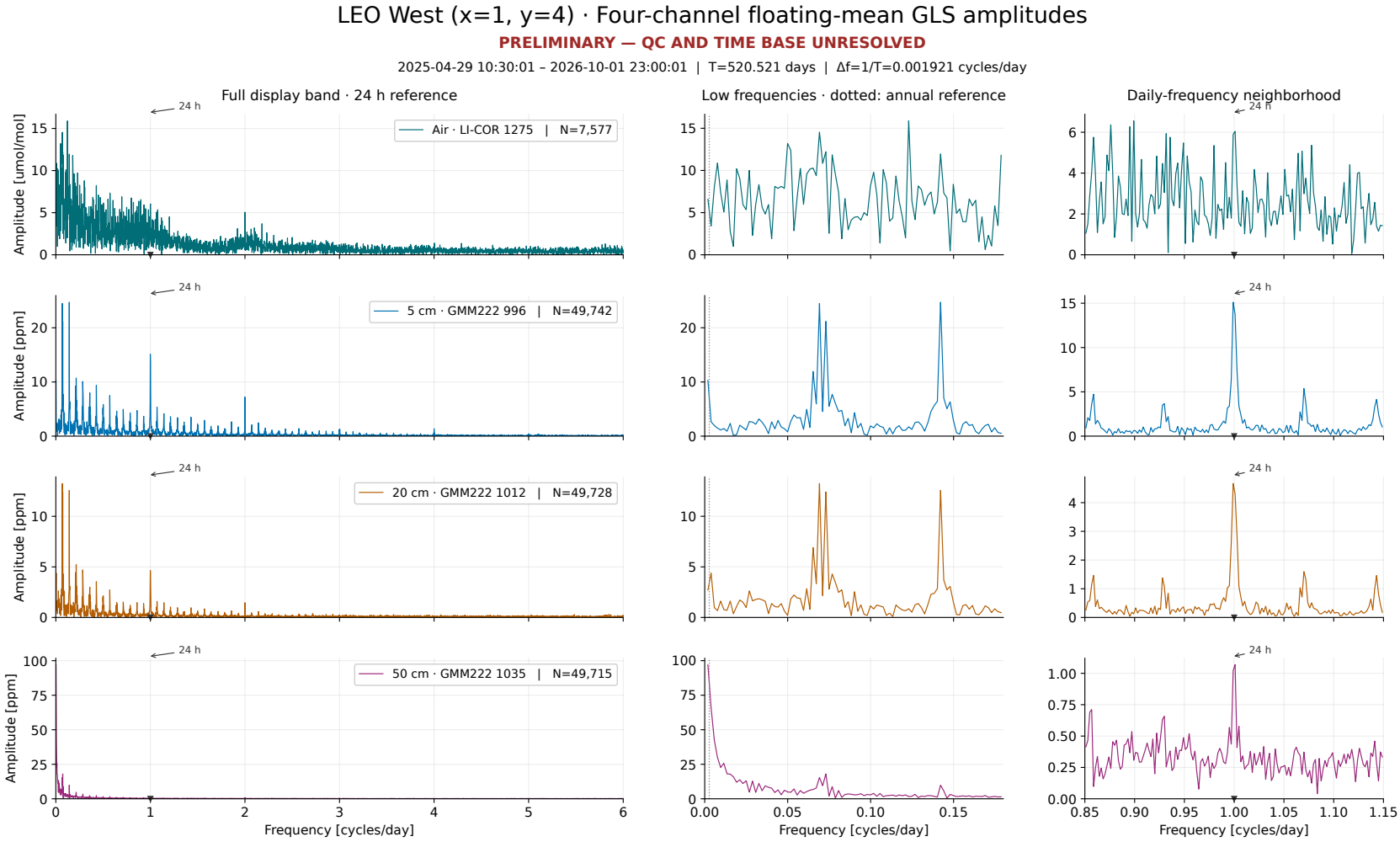


Figure 1: Pre-existing four-channel preliminary amplitude spectra, obtained by independent floating-mean sinusoidal fits. A daily-frequency marker does not constitute frequency selection or phase validation.

4.2 Stage 0.4: direct complex coefficients and relative phases

Set t_{ref} =29 April 2025 00:00. For each common two-hour center $t_k = s_k + 1$ hour, express $t_k - t_{\text{ref}}$ in days. The working frequency is $f_0 = 1$ cycle/day, window length 30 days, and step one day. For channel j use

$$\bar{x}_{j,W} = \frac{1}{N} \sum_{k \in I_W} x_{jk}, \tag{4}$$

$$F_j(f_0; W) = \frac{1}{N} \sum_{k \in I_W} (x_{jk} - \bar{x}_{j,W})\, e^{-2\pi i f_0 t_k}. \tag{5}$$

Here t_k is understood relative to the common time origin. Each occupied two-hour cell receives equal weight regardless of its raw observation count. No additional detrending or taper is applied. This is a direct complex Fourier sum on the actual common timestamps, not a gap-compressed FFT or a GLS fit. For ordered neighboring levels $u \to v$ define

$$Q_{uv} = F_u \overline{F_v}, \qquad \Delta\phi_{u \to v} = \text{wrap}(\text{Arg}\, Q_{uv}), \tag{6}$$

$$\text{wrap}(\alpha) = (\alpha + \pi) \bmod (2\pi) - \pi \in [-\pi, \pi). \tag{7}$$

Thus $\Delta\phi_{\text{air}\to 5}$, $\Delta\phi_{5\to 20}$ and $\Delta\phi_{20\to 50}$ are the report's three principal observables. For ideal sinusoidal pairs $u(t) = \cos(\omega_0 t)$ and $v(t) = \cos(\omega_0(t - \tau))$ on a complete balanced grid, where $\omega_0 = 2\pi f_0$, one has $F_u \propto 1$ and $F_v \propto e^{-i\omega_0\tau}$; hence

$$\Delta\phi_{u \to v} = \text{wrap}(\omega_0 \tau). \tag{8}$$

A positive phase denotes a lag of the second *ideal sinusoid*, modulo 24 hours. This identity does not establish a gas-transport lag or transport direction in LEO.

4.3 Computational implementation and validation

The existing `Project_description/sensorDB/preliminary_fourier_phases.py` defines `aggregate_bins(frames,` `direct_fourier(t_days,x,f0=F0)`, `phase_difference(f_upper,f_lower)`, `wrap(phi)` and `compute_windows(gr` The actual implementation uses `pandas.Series.dt.floor`, `DataFrame.groupby(...).agg`, `reindex`, and `numpy.mean`, `numpy.exp`, `numpy.sum`, `numpy.conj`, `numpy.angle`. This is a verifiable mapping to existing Python methods, not a new listing.

The saved synthetic checks recover zero for identical signals, $+\pi/3$ for a four-hour delay of the second sinusoid, $-\pi/3$ when channels are reversed and $-5\pi/6$ for a 14-hour delay wrapped from $7\pi/6$. The selected irregular common-bin mask gives 1.0515280496203037 rad rather than $\pi/3$, an error of 0.004330498423706031 rad in that synthetic case. On a complete synthetic grid, the maximum complex difference from a correctly phase-corrected `numpy.fft.fft` result was 1.9421×10^{-15}. The original validation recorded 13 passing tests; the later Q1 diagnostic recorded 19 passing tests in total. These test results verify arithmetic and sign conventions, not physical sensor reliability [4, 6, 7].

5 Preliminary results

5.1 Three temporal series

The saved stage 0.4 run reports 492 sliding 30-day windows, with all three relative phases numerically defined in every window. Their centers run from 14 May 2025 to 17 September 2026. Every window retains unresolved review flags, and a scientific reliability classification has *not* been established. A count of zero scientifically certified windows here indicates lack of certification by this procedure, not evidence that every window is invalid.

The three angular coordinates define a candidate point in $\mathbb{T}^3 = S^1 \times S^1 \times S^1$. The cube $[-\pi, \pi)^3$ is only one coordinate display with opposite faces identified; a plotted branch jump or Euclidean cube distance is not automatically intrinsic motion on the torus.

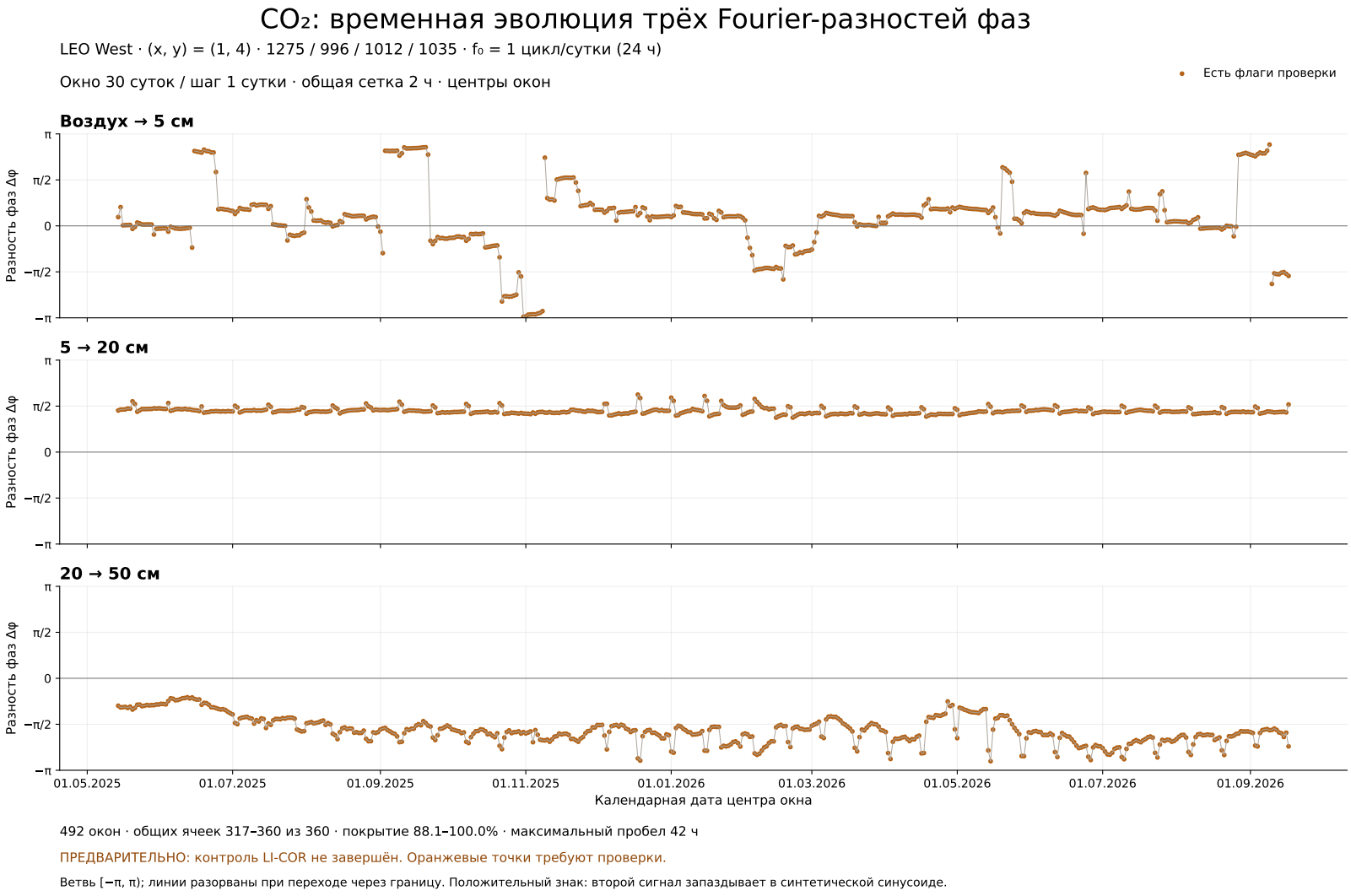


Figure 2: Pre-existing time evolution of three direct-Fourier relative phases, 30-day windows, one-day step, one cycle/day. All windows carry review flags.

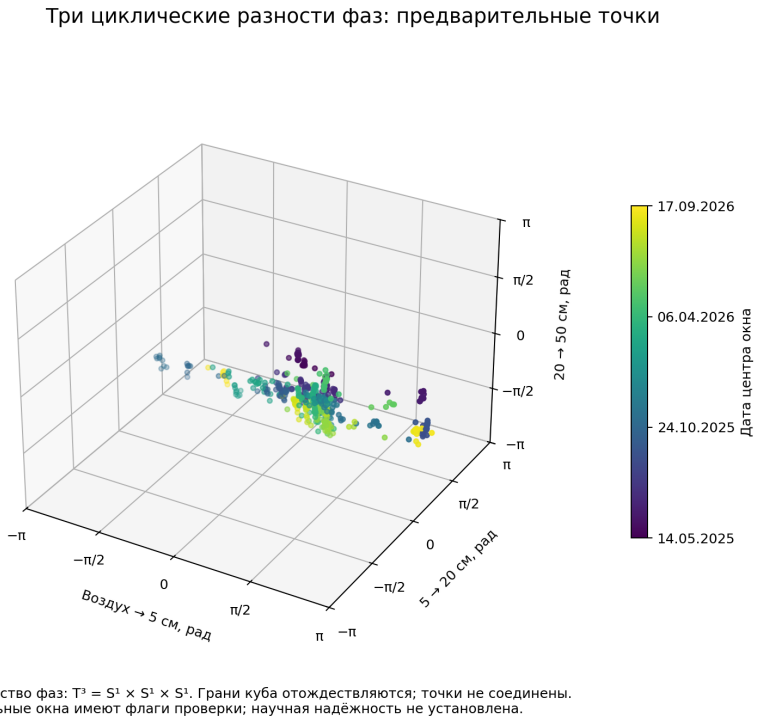


Figure 3: Pre-existing visualization of the three cyclic phase coordinates; it is not a demonstrated physical trajectory or a validated topological invariant.

5.2 One 5-to-20 cm discontinuity examined for Stage 0.5/Q1

The latest available diagnostic selects a single adjacent pair by maximal absolute difference on the original $[-\pi, \pi)$ phase branch, with earlier occurrence breaking ties. The centers are 17 and 18 December 2025, corresponding to 30-day windows from 2 December through 1 January and from 3 December through 2 January, respectively. Neighboring windows overlap for 29 days and must not be treated as independent events. For the complex product $Q = F_5\overline{F_{20}}$, distinguish

$$d_Q = \mathrm{wrap}(\mathrm{Arg}\, Q_2) - \mathrm{wrap}(\mathrm{Arg}\, Q_1), \qquad \delta_Q = \mathrm{wrap}(\mathrm{Arg}(Q_2\overline{Q_1})), \qquad d_Q - \delta_Q = 2\pi k,\ k \in \mathbb{Z}. \tag{9}$$

The saved phases are 1.3815969902 and 1.9625409175 rad, so $d_Q = \delta_Q = +0.5809439273$ rad and $k = 0$: *this change is not an argument-branch crossing.* The coefficient argument changes are +0.0891610601 rad for F_5 and −0.4917828672 rad for F_{20}, agreeing with the product-phase increment. Their moduli decrease from 10.087605 to 6.749558 ppm and from 2.715018 to 2.004241 ppm. All eight neighboring windows have 360/360 common bins, but all contain review flags. None of these sampled coefficients is zero. There is no established amplitude-error threshold and no knowledge of an intervening continuous coefficient path, so a zero crossing cannot be proved or excluded. No physical event or mechanism is established [4, 7].

6 Historical analyses, not spectral-phase evidence

The earlier Chapter 1 fitted a fixed diurnal harmonic. Earlier Chapter 2 studied the different West vertical W_R4_C-4 from 11 September 2013 through 30 September 2026, over 157 months and 2,614,638

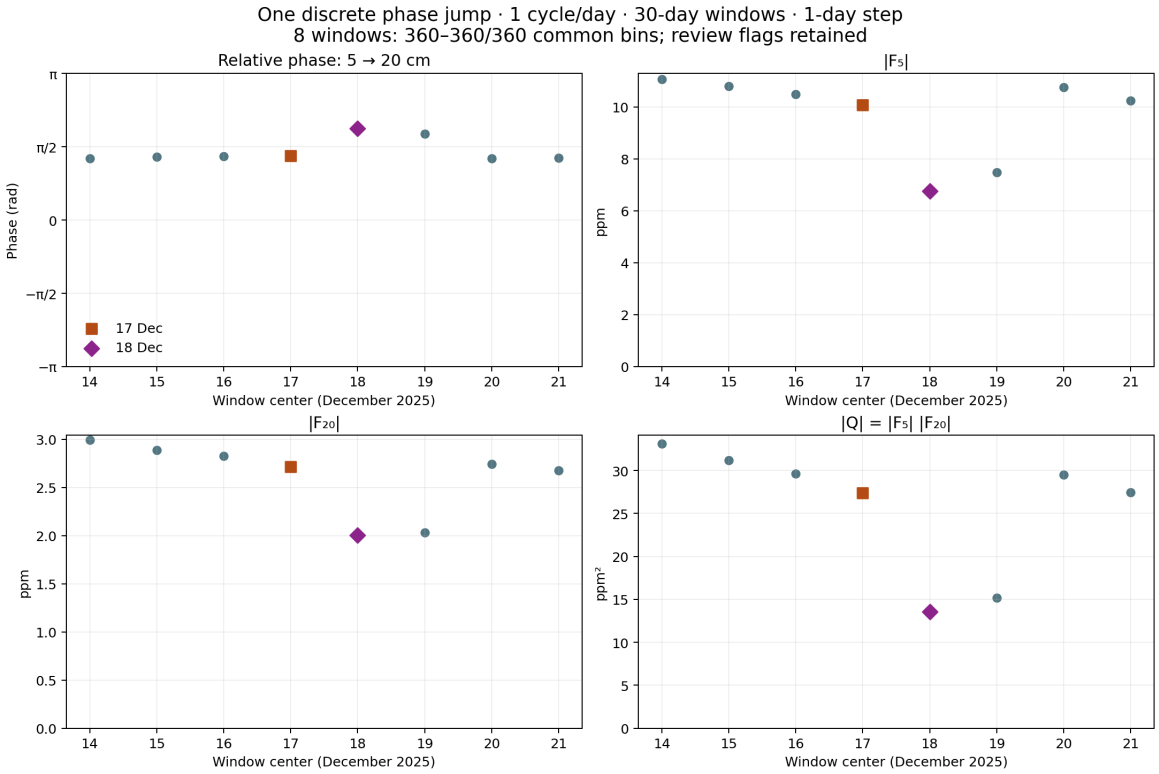


Figure 4: Saved discrete local relative phase and coefficient moduli around one selected phase change.

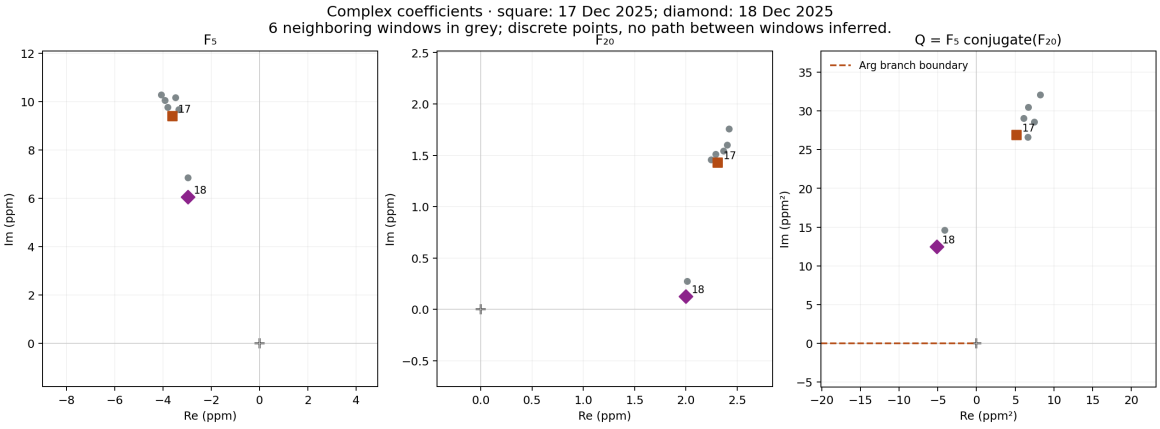


Figure 5: Saved complex Fourier coefficients and their product Q, showing the origin and the relevant argument branch boundary without interpolating an unknown path.

archived raw measurements. Its final historical processing retained 43 jointly admitted months, 23 vectors of absolute daily maximum times $H(d)$ and 25 daily-amplitude vectors $A(d)$. These are not four-channel spectral-window admissions. Earlier Chapter 3 found no established reproducible long-term evolution under its original absolute-maximum-time and amplitude procedure. The historical input and its limitations are retained but are not recalculated or interpreted as either confirming or refuting the new relative spectral-phase problem [1, 9].

7 Mathematical validation and physical interpretation

A Fourier phase is defined only for nonzero complex coefficients; when a modulus approaches zero, the argument becomes sensitive to perturbations. A discrete branch crossing must be distinguished from an intrinsic circular increment by equation (9). Missing samples and changing source quality can affect direct Fourier coefficients. The successful synthetic and Python/browser cross-checks certify implementation properties, not error bars or a physical causal model.

A reduced diffusion–reaction reference equation might be written

$$\partial_t C = \partial_z(D_{\text{eff}}\,\partial_z C) - \partial_z(vC) + S - R, \tag{10}$$

but it has not been fitted or uniquely identified in this pilot. Moisture, storage, boundary forcing, thermal variations, physical sensor quality, and source/sink terms remain possible confounders. Phase alone is not a diffusion coefficient and cannot identify biological versus geochemical origin [2, 13].

8 Buchstaber framework: approved methodological preparation only

The separate existing Chapter 4 notebook states that application to LEO has not begun. For an admissible uniform scalar series $(f_1, \ldots, f_N)$, the proposed unfolding consists of

$$X_k = (f_k, \ldots, f_{k+n-1})^{\mathsf{T}} \in \mathbb{R}^n, \quad P = N - n + 1, \quad k = 1, \ldots, P. \tag{11}$$

Writing $\bar{X} = P^{-1}\sum_k X_k$ and $S = P^{-1}\sum_k (X_k - \bar{X})(X_k - \bar{X})^{\mathsf{T}}$, with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$, the quadratic affine projection identity is

$$E_q = \min_{\dim L = q} \frac{1}{P}\sum_k \|(I - \Pi_L)(X_k - \bar{X})\|_2^2 = \sum_{i=q+1}^{n} \lambda_i, \qquad L_q^* \in G(n, q). \tag{12}$$

This standard mathematical identity is not a measured LEO result. In particular the Grassmannian $G(n, q)$ is not the phase torus $\mathbb{T}^3$. Synthetic tests, validated data and a separately approved protocol are prerequisites for application to LEO [10, 11, 12].

9 Limitations and open questions

The scientific admission of LI-COR and basalt signals is unfinished. No confirmed threshold for phase stability or amplitude has been selected. Overlapping windows are dependent; direct Fourier coefficients may be affected by missingness and spectral leakage. The adopted daily frequency is provisional; investigation of phase as a function of frequency is not currently authorized. Complete actual West-slope irrigation events (including SCADA logs, timing, intensity and location) remain outstanding. The public Arevalo et al. data provide only partial 2015–2019 precipitation coverage and cannot replace the complete operational history. None of the preliminary curves identifies a chemical, biological, or transport mechanism [2].

10 Next approved steps

The existing PLAN requires documented spectral-specific quality and coverage, the actual irrigation calendar or an explicit decision about its missing portion, presentation of the four amplitude spectra and three temporal phase curves with formulas and tests, and explicit feedback from I. R. Gabitov on frequency, time window, step and experimental interval. Until those completion conditions hold, the new spectral Chapter 1 cannot begin. The single Q1 discontinuity diagnostic does not authorize wider jump analysis or mechanism inference. The separately allowed Buchstaber preparation remains subject to its own input and approval conditions [2].

Sources, figures and reproducibility

This report consolidates the source repository at commit `67aa98954162d0437bfcc30d7d7b14c4d531f10f`, 19 October 2026. The existing figures used here, without regeneration, are:

- `output/preliminary_spectra/four_channel_spectra_17x11.pdf`;
- `output/preliminary_fourier_phases/phase_evolution_17x11.pdf`;
- `output/preliminary_fourier_phases/phase_trajectory_3d.png`;
- `output/preliminary_fourier_phases/jump_diagnostic/phase_jump_local.png`;
- `output/preliminary_fourier_phases/jump_diagnostic/phase_jump_complex.png`.

The original numerical and audit files remain in `output/preliminary_spectra/` and `output/preliminary_fourier_` as passport, validation, phase, coverage and Q1 diagnostic tables. Supplementary interactive HTML is `output/preliminary_spectra/interactive_spectra.html` and `output/preliminary_fourier_phases/phase_e` Compile this sole master file from `Project_description/Research_log/` using `latexmk -pdf -interaction=nonst -halt-on-error Research_Report.tex`.

References

[1] CO2Flux, *Project_description/Research_log/README.md*, revision 4 October 2026.

[2] CO2Flux, *Project_description/Research_log/PLAN.md*, preparatory Stage 0 and unified report policy, source commit specified above.

[3] CO2Flux, *00_preliminary_co2_spectra.ipynb*, and *output/preliminary_spectra/passport.json*.

[4] CO2Flux, *00_preliminary_fourier_phases.ipynb*, including Section 10 (single Q1 phase jump).

[5] CO2Flux, *output/preliminary_fourier_phases/passport.json*.

[6] CO2Flux, *output/preliminary_fourier_phases/validation.json*.

[7] CO2Flux, *output/preliminary_fourier_phases/jump_diagnostic/phase_jump_validation.json*.

[8] CO2Flux, *details/02_data_checks.md*, verified computational procedures and stated limitations.

[9] CO2Flux, *01_october_2017.ipynb; 02_monthly_changes.ipynb; 03_long_term_evolution.ipynb*.

[10] CO2Flux, *04_buchstaber.ipynb*, proposed method only.

[11] V. M. Buchstaber, *Time Series Analysis and Grassmannians*, American Mathematical Society Translations, Series 2, vol. 162 (1994), pp. 1–17.

[12] V. M. Buchstaber, *Kolmogorov Widths, Grassmann Manifolds and Unfoldings of Time Series*, Matematicheskii Sbornik, vol. 216, no. 3 (2025), doi: 10.4213/sm10240.

[13] M. Maier and H. Schack-Kirchner, *Using the gradient method to determine soil gas flux: A review*, Agricultural and Forest Meteorology 192–193 (2014), 78–95, doi: 10.1016/j.agrformet.2014.03.006.